D.S. L-1.

1. Proposition => [Statement] A proposition is a declarative sentence which is either true or false => truth value.

examples => i> 1+1 = 2 True
ii> 9<6 False
iii> What is this ?
iv> Read this.] -> Not statement.

Compound Proposition => When one or more purepositions are connected through various connectivities is called compound proposition.
by using and, or
ex => I am eating my food 'and' I am going to college.

-> A purposition is said to be primitive if it cannot be broken down in to simpler purposition.
ex => I am eating ~~ap~~ an apple.

Basic Logical operations ->

1. Conjuction -> AND -> '$\wedge$'
2. Disjuction -> OR -> '$\vee$'
3. Negation -> NOT -> '$\sim$'
4. NAND -> NOT + AND
5. NOR -> NOT + OR
6. R.

1. **Conjuction** ⇒ Any two purupositions can be combined by the word 'and' to form a compound purupasition called 'conjuction'.

Let → P & q ⇒ two puruposition
$(P \wedge Q)$ → Conjuction

Truth Table

P	q	$P \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

2. **Disjunction** ⇒ Any two puruposition combined by the word 'or' is called disjunction.

$(P \vee q)$ → disjunction.

Truth table.

P	q	$P \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

3. **Negation** ⇒ A single puruposition which value can be changed by using Negation.

$\sim P$, $\sim q$

P	$\sim P$
T	F
F	T

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L-2 discrete S.

Tautologies & Contradiction =)

1. **Tautology** =) Truth values are true for any truth value of variable.

=) In the last column if all value are true.

2. **Contradiction** =) Truth values are false for any truth value of variables.

=) when all values in last column are false.

3. **Contigency** =) Some truth values are true and some are false.

ex i) $P \vee Q$.

P	Q	$P \vee Q$
T	T	T
T	F	T
F	T	F
F	F	F

Not Tauto.
Contigency

ii) $P \wedge \sim P$

P	$\sim P$	$P \wedge \sim P$
T	F	F
F	T	F

Contrad.

iii) $P \vee \sim P$

P	$\sim P$	$P \vee \sim P$
T	F	T
F	T	T

Taut.

iv) $\sim (P \wedge Q)$

Logical equivalence =) Two propositions are said to be logically equivalence or equal if they have same truth values.

=) If last column of two statements are exactly same

→ Denoted as $P \equiv Q$ (if P & Q are two propositions)

ex i) $\sim(P \vee q)$

P	q	$(P \vee q)$	$\sim(P \vee q)$
T	T	T	F
T	F	T	F
F	T	T	F
F	F	F	T.

ii) $\sim P \wedge \sim q$

P	$\sim P$	q	$\sim q$	$\sim P \wedge \sim q$
T	F	T	F	F
T	F	F	T	F
F	T	T	F	F
F	T	F	T	T

So we can say $\sim(P \vee q) \equiv \sim P \wedge \sim q$

Conditional Statement ⇒

→ Statements in the form of "if P then q"

→ Denoted by $P \rightarrow q$.

→ The conditional statement is false when 'P' is true and 'q' is false, and true otw.

→ P is called 'hypothesis' and q is called 'Conclusion'

ex " If I am elected, then I will lower taxes".
If the politician is elected, voters would expect this politician to lower the taxes. Furthermore, if the politician is not elected, then voters will not have any expectation ~~that~~ this person will lower taxes. ~~Although~~ It is only when the politician is elected but does

mat lower taxes that voters can say that the politician has broken the campaign pledge. The last scenario corresponds to the case when p is true but q is false ~~in~~ in $p \rightarrow q$.

Truth table

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Biconditional statement ⇒

- ⇒ Statement in the form of "p if and only if q".
- ⇒ denoted by $p \leftrightarrow q$.
- ⇒ Biconditional is true when p and q have the same truth values and false otw.
- ⇒ Biconditional statements are also called bi-implications

☆ "p is necessary and sufficient for q".
☆ "If p then q, and conversely".
☆ "p iff q".

Ex "You can take a flight ~~iff~~ if you buy a ticket".
⇒ This statement is true if p and q are either both are true ~~an~~ or both are false. That is, if you buy a ticket can take a ~~flight~~ flight.

It is false when p and q have opposite truth values, that is ~~case~~ "when you do not buy a ticket, but you can take the flight. and when you buy a ticket but you can't take the flight.

Truth table

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Inverse, Converse and Contrapositive :)

$$P \rightarrow Q.$$

Inverse => $\sim p \rightarrow \sim Q$

Converse => $Q \rightarrow P$

Contrapositive => $\sim Q \rightarrow \sim P$

ex "The home team wins whenever it is raining".

Original statement => "If it is raining, then the home team wins".

|||

Contrapositive => If the home team does not win, then it is not raining.

Converse => If the home team wins, then it is raining.

Inverse => If it is not raining then home team does not win.

L-3 9/8/2021

Predicates and Quantifiers.

Predicates ⇒ Propositional logics cannot express the meaning of all the statements in mathematics and natural language.

ex $x>3$, $x=y+3$, $x+y=z$.

ex Computer x is functioning properly.

Statements involving variables are neither true nor false when values of statements are not specified.

ex 1. $x>3$.

This example has two parts first is the subject of the statement and second is the predicate "is greater than 3" ⇒ refers to a property that the subject of statement can have.

So $P(x) = x>3$. ⇒ the value of propositional function P a x.

Note.

Q. Let $P(x)$ denotes the statement "$x>3$", what are the value of $P(4)$ and $P(2)$.

→ $P(4)$ ⇒ True, $P(2)$ ⇒ false.

ex Let $Q(x,y)$ denote the statement "$x=y+3$"

$Q(1,2)$ and $Q(3,0)$?

ex $R(x,y,z)$ denotes $x+y=z$

$R(1,2,3)$ & $R(0,0,1)$?

ex $R(x,y,z)$ denotes ~~between~~ $7x-4y=z$

$(1,1,3)$ $(2,4,5)$, $(6,1,7)$ = ?

Note ⇒ A statement of the form $P(x_1, x_2, x_3 - - - x_n)$ is the value of Propositional function P at the n-tuple $(x_1, x_2 - - x_n)$ and P ⇒ n-place predicate or a n-ary predicate.

Preconditions and Postconditions ⇒

Precondition ⇒ The statement that describe the valid input

Post Conditions ⇒ ~~The~~ ~~states~~ the conditions that the output should ~~satis~~ ~~Sta~~ satisfy when program has run is Post Conditions

Quantifiers

Quantification express the extent to which a predicate is true over a range of elements.

In English ⇒ All, some, many, none, few.

(a) Universal quantification ⇒ A predicate is true for every element under consideration.

(b) Existential quantification ⇒ One or more element.

ex ~~Universal~~ (a) Some men are clever.

(b) All ~~the~~ graphs are connected.

(c) Some dogs are white.

Universal quantifiers ⇒ The universal quantification of $P(x)$ is the statement

"$P(x)$ for all values of x in the domain".

Notation ⇒ $\forall x \; P(x)$

$\forall$ ⇒ universal quantifiers.

An element for which $P(x)$ is false is a counterexample of $\forall x\, P(x)$.

Statement	When true	When false.
$\forall x\, P(x)$	$P(x)$ is true for every x	There is an 'x' for which $P(x)$ is false.
$\exists x, P(x)$	There is an 'x' for which $P(x)$ is true	$P(x)$ is false for every x.

ex $P(x) = x+1 > x$. domain ⇒ all real numbers

ex $Q(x) = x < 21$ domain ⇒ all real numbers.

$x = 22$

ex $\forall x\, P(x)$ $x^2 < 10$ domain Z^+ not exceeding 4.

Existential quantifiers :) The existential quantification of $P(x)$ is the proposition

"There exists an element x in the domain such that $P(x)$".

$\exists x\, P(x)$ ⇒ existential quantification, $\exists$ ⇒ existential quantifier.

ex $\exists x\, P(x)$, "$x > 3$" domain ⇒ all real numbers.

ex $\exists x\, Q(x)$, $Q(x) = x = x+1$ domain ⇒ all real numbers

Logic Programming. - S2

Nested Quantifiers ⇒ Where one quantifier is with in scope of another.

ex $\forall x \exists y (x+y=0)$

this is same thing as $\forall x\, Q(x)$; where $Q(x)$ is $\exists y\, P(x,y)$
where $P(x,y)$ is $x+y=0$

ex $\forall x \forall y (x+y=y+x)$ domain $\mathbb{R}$

ex $\forall x \exists y (x+y=0)$ $d \Rightarrow \mathbb{R}$

ex $\forall x \forall y ((x>0) \text{ and } (y<0) \rightarrow (xy<0))$ $D \Rightarrow \mathbb{R}$

The order of Quantifiers ⇒

ex $P(x,y) = x+y = y+x$ $D=\mathbb{R}$

$\forall x \forall y\, P(x,y)$ OR $\forall y \forall x\, P(x,y)$ which are True.

⇒ Nested universal quantifiers in a statement without other quantifiers can be changed without changing the meaning of the quantified statement.

ex $Q(x,y)$ denotes $x+y=0$
Truth values for $\underset{F}{\exists y \forall x\, Q(x,y)}$ and $\underset{T}{\forall x \exists y\, Q(x,y)}$; $D=\mathbb{R}$

⇒ a) $\exists y \forall x\, Q(x,y)$ ⇒ There exist single value of y such that for every x $x+y=0$

b) $\forall x \exists y\, Q(x,y)$ ⇒ for every x, $\exists y$ such that $x+y=0$ ⇒ $x=-y$

Statement	When true	When false.
a) $\forall x \forall y\, P(x,y)$ $\equiv$ $\forall x \forall x$	$P(x,y)$ is True for every (x, y)	There is pair x, y for which $P(x, y)$ is false.
b) $\forall y \forall x\, P(x,y)$	"	"
c) $\forall x \exists y\, P(x,y)$	For every x, there exist y for which $P(x,y)$ is true	There is an x such that $P(x,y)$ is false for every x.
d) $\exists x \forall y\, P(x,y)$	There is an x for which $P(x,y)$ is true for every y.	For every x there is a y for which $P(x,y)$ is false.
e) $\exists x \exists y\, P(x,y)$ $\exists y \exists x\, P(x, y)$	There is pair of x, y for which $P(x, y)$ is true	$P(x, y)$ is false for every pair x, y).

Unit-2

Proof ⇒ A Proof is a valid argument that establishes the truth of a mathematical statement.

eg. If $x > y$ where x, y are +ive real numbers, then $x^2 > y^2$.

Proving theorems ⇒

a) **Direct Proof** ⇒ An implication $P \to q$ can be proved by showing that if P is true then q is also true.

Ex If n is odd, then n^2 is odd.

$n = 2k+1$, $n^2 = (2k+1)^2 \Rightarrow 4k^2+4k+1$

$= 2(2k^2+2k)+1$

let $2k = m$ $= 2(2m+1)+1$

If we take any value of m we get $n^2 =$ odd.

Ex The Product of two odd integers is odd.

$a = 2m+1$, $b = 2n+1$

$(2m+1)\cdot(2n+1) = 4mn+2m+2n+1 = 2(2mn+m+n)+1$

$\Rightarrow$ let $2mn+m+n = k$

$2k+1 \Rightarrow$ odd.

If we treat $2mn+m+n$ as single integer k we have $2k+1$ which is also odd.

b) Indirect Proof (Contraposition):-

An implication $p \to q$ is equivalent to its contra-position $\neg q \to \neg p$ $\Rightarrow$ whenever q is false, q is also false.

Ex If $\underline{3n+2 \text{ is odd}}_{p}$, then $\underline{n \text{ is odd}}_{q}$.

n is even $\Rightarrow$ $3n+2 \Rightarrow 3(2k)+2 = 6k+2$

$= 2(3k+1) = \underbrace{2\cdot m}_{\text{even}}$ let $m = 3k+1$

$\neg q \to \neg p$ so $p \to q$.